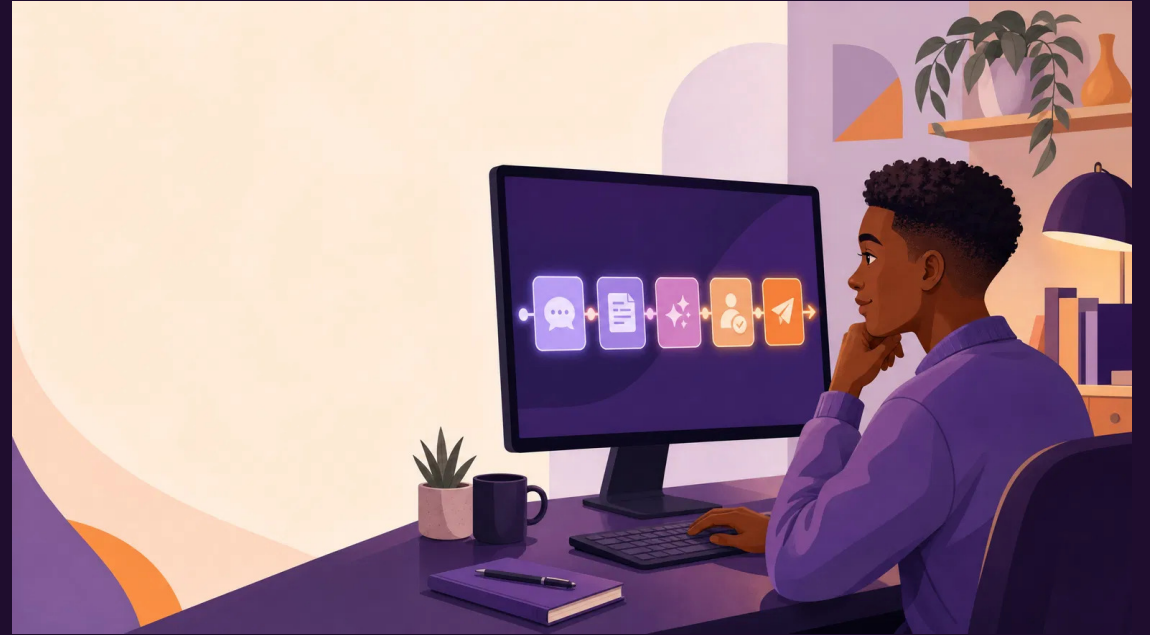


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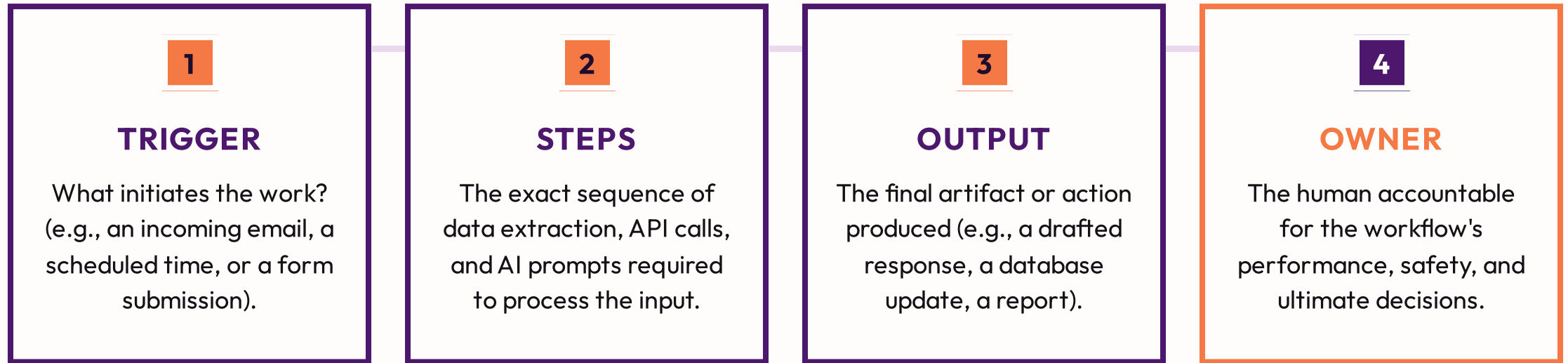
AI Workflows & Automation

Move beyond single prompts. Learn how to orchestrate models, tools, and human oversight into reliable automated systems.

YOU ARE THE ORCHESTRATOR



From Task to Workflow



The shift: Instead of prompting an AI every time you need something, you build a reliable machine that runs the prompts for you when specific conditions are met.

Workflow or Agent?

AI Workflow

- **Controlled Sequence:** Orchestrates models and tools through predefined code paths.
- **Predictable:** Executes the exact same steps in the exact same order every time.
- **Best for:** Well-defined, repeatable tasks with clear inputs and outputs.

AI Agent

- **Dynamic Decisions:** The LLM directs its own process, choosing which tools to use and when.
- **Flexible:** Adapts to new information and recovers from errors autonomously.
- **Best for:** Open-ended problems where the required steps cannot be hardcoded.

The Rule: Always use the simplest solution possible. Build a workflow first, and only introduce an agent when the task genuinely requires autonomous planning.

Automation Candidates

1. REPEATABLE

The task occurs frequently enough that the time saved by automating it outweighs the initial setup and maintenance costs.

Example: Weekly status reports

2. STRUCTURED

The inputs follow a predictable format, and the rules for processing them are clear, consistent, and easily defined.

Example: Standardized intake forms

3. LOW-RISK

Errors will not cause catastrophic harm, or the workflow includes mandatory human oversight before final execution.

Example: Drafting internal emails

BEST PRACTICE:

Always start by automating a simple, low-risk, highly repeatable task before attempting complex, multi-step processes.

The Core Pattern

01 TRIGGER

The event that starts the process, such as receiving an email, a webhook, or a scheduled time.

02 CONTEXT

Gathering necessary background data from databases or APIs to inform the AI's decision.

03 AI STEP

Prompting the language model with the context to generate a draft, classification, or decision.

04 GATE

A validation check or human approval step to ensure the AI's output is safe and accurate.

05 ACTION

Executing the final task, such as sending a message, updating a record, or triggering another system.

06 LOG

Recording the inputs, outputs, and outcomes for future auditing, debugging, and improvement.

Prompt Pipelines

Instead of one massive prompt that tries to do everything, break the task into a chain of smaller, specialized prompts.

1. Extraction

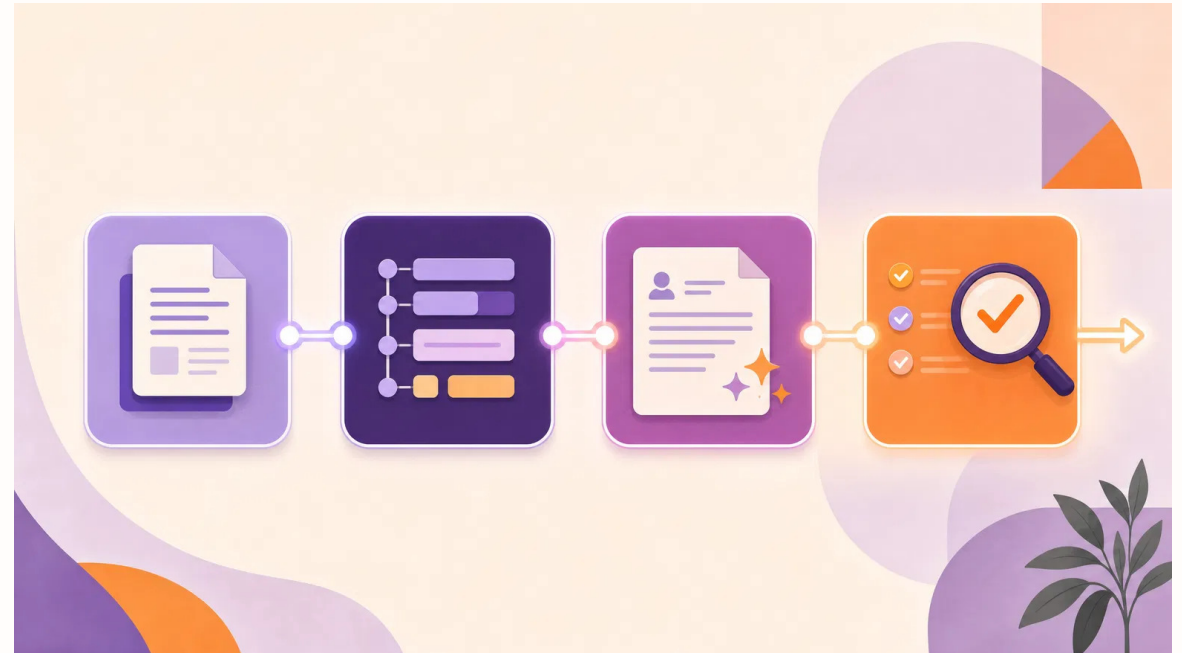
Pull specific facts, entities, or data points from raw input.

2. Drafting

Write content based strictly on the extracted facts.

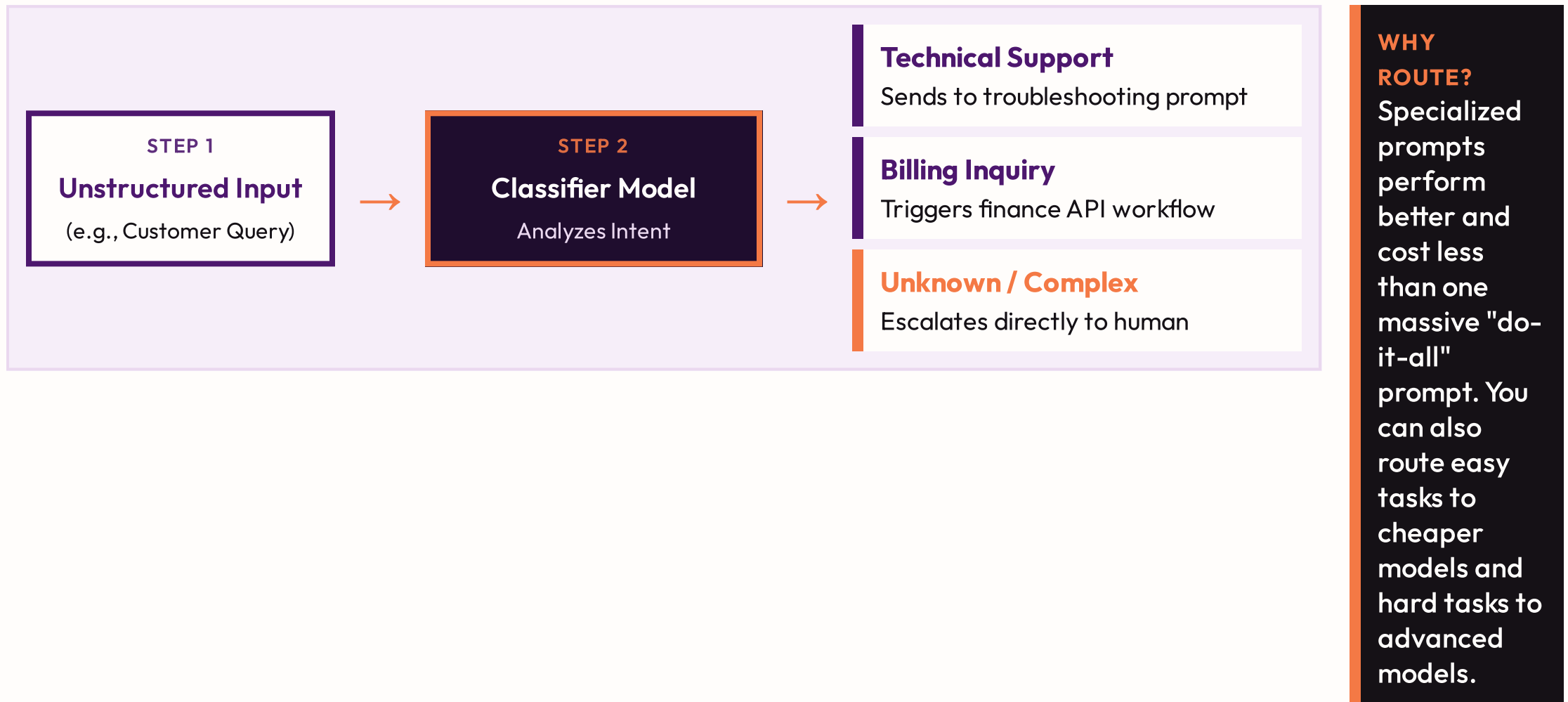
3. Quality Check

Evaluate the draft against constraints before finalizing.



Routing

Routing classifies an incoming request and directs it to a specialized downstream task. This separates concerns and allows you to build highly focused prompts for specific scenarios.



Human-in-the-Loop

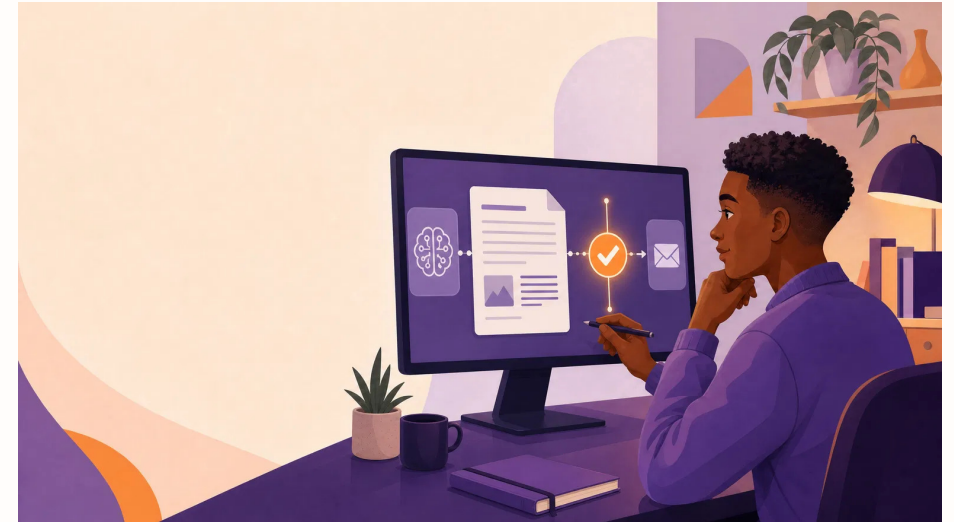
Automation does not mean removing humans. It means shifting humans from creators to reviewers and decision-makers.

APPROVAL CHECKPOINTS

The AI completes the heavy lifting (drafting, analyzing, extracting), but the workflow pauses. A human must explicitly approve the output before the final action is executed.

ESCALATION PATHS

If the AI encounters an unknown scenario, low confidence score, or high-risk trigger, the workflow automatically routes the task to a human expert.



Data Boundaries

Protect sensitive information by strictly controlling what data enters and exits your automated workflows.

Least Disclosure

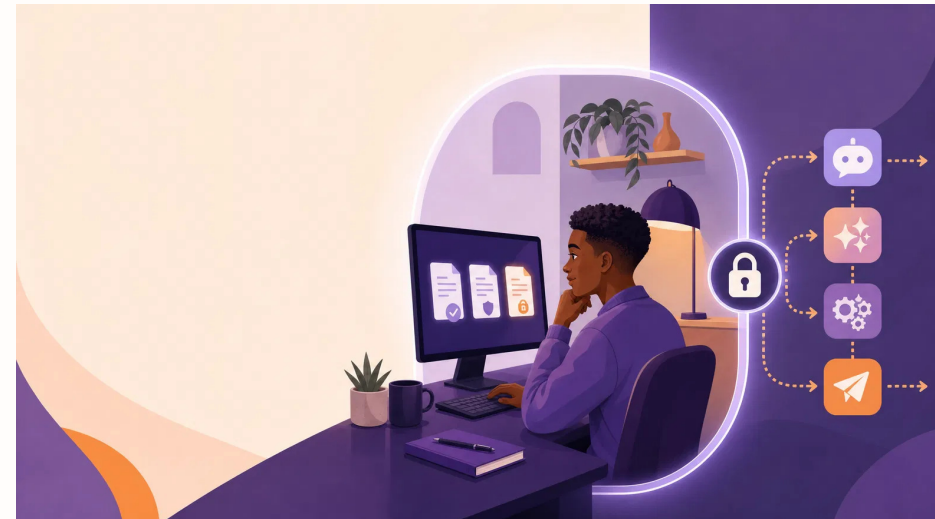
Only pass the exact data fields the AI needs to complete its specific task, never the entire database record.

Access Controls

Ensure the workflow runs with restricted permissions, preventing it from accessing or modifying unauthorized systems.

Privacy Scrubbing

Automatically redact Personally Identifiable Information (PII) before sending data to external language models.



Testing a Workflow

1. THE HAPPY PATH

Confirm the workflow succeeds with ideal inputs to validate core logic and integrations.

Focus: Works when everything is correct.

2. EDGE CASES

Test unusual or malformed inputs to check safety and ambiguity handling.

Focus: Fails safely under odd inputs.

3. FAILURE RECOVERY

Simulate API timeouts and other failures to ensure graceful recovery and alerts.

Focus: Alerts humans when intervention is needed.

THE REALITY: Automated workflows must be tested for failures before deployment.

REVIEW

Knowledge Check

Scenario 01: A team wants to automate writing personalized performance reviews for employees based on raw peer feedback notes.

**HIGH RISK
(NEEDS HUMAN
GATE)**

Scenario 02: A company wants to extract invoice numbers, dates, and total amounts from standard PDF receipts submitted by vendors.

**GOOD WORKFLOW
CANDIDATE**

Scenario 03: A customer service bot needs to dynamically decide whether to search the web, query a database, or ask the user for clarification based on the conversation flow.

**AGENT
(NOT WORKFLOW)**

Lab Launch: Build an AI Workflow

Design and deploy a simple automated workflow (using a low-code tool or script) to process incoming study materials, categorize them, and draft a summary.

— DELIVERABLES

- **Trigger & Context:** A defined event that starts the workflow and gathers necessary data.
- **Prompt Pipeline:** At least two chained prompts (e.g., one for extraction, one for drafting).
- **Human Gate:** A checkpoint where you must approve the draft before it is saved or sent.

— SUCCESS CRITERIA

- The workflow runs end-to-end without crashing on the happy path.
- Data boundaries are respected (no sensitive personal data is passed to the LLM).
- The system fails safely and alerts you when given unexpected input.

EXTENSION CHALLENGE

Implement a **Routing** step. Use a lightweight classifier prompt to determine if the input is "Technical" or "General", and send it down two different processing paths.